

Machine Learning for Modelling: *Supervised Learning*

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Exam Project

Valid for June/July/Sept 2026 and Jan/Feb 2027 sessions

Task and dataset



- **Task:** Image classification using CNNs (Supervised Learning) and compare the performance with CNNs trained in Self-Supervised Learning (even on the same dataset ignoring the labels), extracting the features and classifying them with a traditional classifier
- **Dataset:** Food recognition
 - uncontrolled input size
 - 251 classes
 - from 100 to 600 images per class
 - Train, Val, Test splits: Test will not be used since it does not have a public ground truth, your test set is the validation set
 - validation set must be extracted from the training set

Task and dataset



- **Additional constraints:**
 - The CNN should be a custom model with a total number of parameters $< 10M$
 - For (motivated and documented) computational reasons, the number of images for each class can be reduced, while the number of classes must remain 251
- Dataset download link: available on the e-learning platform

Tasks



1. Data Preprocessing: The students should preprocess the images by resizing them to a common size, converting them to grayscale or RGB, and normalizing the pixel values.
2. Convolutional Neural Networks: The students should train a convolutional neural network (CNN) on the preprocessed images to classify them into different categories. The student should experiment with a custom neural architecture.
3. The students have to cast the problem as supervised learning (SL) and self-supervised learning (SSL).
4. Model Evaluation: The students should evaluate the performance of the models using metrics such as accuracy, precision, recall, and F1-score.
5. Hyperparameter Tuning: The students should tune the hyperparameters of the models to achieve better performance.

The project must include a written report that describes each step of the project in detail, presents the results, and discusses the limitations and future directions of the project. The report should also include visualizations to support the analysis.

What must be delivered

- All Python software codes associated with the tasks. Use the standard practice you learnt during the lab lectures to describe and comment the software code.
- A technical report consisting of no more than 8 pages (references excluded) where you describe how you analyzed the given dataset, your design choices, your results, etc.
The template to be used is available on the course page in the “Exam” section

What is allowed

- You can develop the project yourself or you can develop the project with one of your colleagues
- You can access any source of high quality to develop your project, other courses, papers, books, ... but it must be properly cited

What is not allowed

- No plagiarism in any form, including chatGPT and similar (you must add a statement to your report that clearly exclude the use of any form of plagiarism)

Deadlines



When it must be delivered

- On Friday before the exam (usually), deadline at 23:59:
 - 19-06-2026 for the 22-06-2026 exam
 - 10-07-2026 for the 13-07-2026 exam
 - 24-07-2026 for the 27-07-2026 exam
 - 18-09-2026 for the 22-09-2025 exam

How it must be delivered

- Write an email to me with the link to a shared folder on Google Drive
- After the deadline no modifications are allowed to the code and the report
- Only (small) changes to the presentation are allowed (e.g., typos correction, etc.)

Assessment



- Ongoing Exam (Max 33 points):

- Lab reports 10 pts
- Project 12 pts
- Interview 11 pts

- Full Exam (Max 30 points):

- Project 20 pts
- Interview 10 pts

Project grading



- **Software code** (up to 6 points)
 - Correctness (up to 2 points)
 - Efficiency (up to 2 points)
 - Clarity of comments (up to 2 points)

- **Technical report** (up to 6 points)
 - Correctness, rigor, accuracy of the methodological approach (up to 2 points)
 - Clarity of expression and communication of ideas (up to 2 points)
 - Tables, figures, captions, are they appropriate? Well motivated? Explained? (up to 2 points)

Interview and rejecting the grade



Interview

— about 20 minutes

- 10 minutes for the oral presentation of the project, slides to be used, both members must present
- 10 minutes of interview on methodological aspects of the project and of what presented in the course

Rejecting the Grade

- A new version of the project, substantially different from the previous one (even if on the same task) must be prepared and submitted
- Lab Reports expire after 1 year (they are valid until February 2026 exam session)
- New interview